

Liouville's Goldbach Problem for Multiples of Four

The problem

For a positive integer n , let $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts prime factors with multiplicity. Thus $\lambda(p) = -1$ for every prime p , and $\lambda(ab) = \lambda(a)\lambda(b)$ for all positive integers a, b .

In 2018, Shusterman asked whether every even integer $N > 2$ can be written as $N = a + b$ with $a, b > 0$ and $\lambda(a) = \lambda(b) = -1$ [1]. This weakens the binary Goldbach conjecture: each summand need only have an odd number of prime factors, counted with multiplicity.

Mangerel proved an unconditional correlation bound [2], recalled below. He later proved Shusterman's assertion for all sufficiently large even integers under the Generalised Riemann Hypothesis for Dirichlet L -functions [3, Theorem 1.2]. The same paper records an elementary unconditional proof for multiples of eight [3, Section 6.2]. We prove the following.

Theorem. Every positive integer divisible by four is the sum of two positive integers whose Liouville values are both -1 .

The only external theorem needed is Mangerel's bound [2, Theorem 1.2]:

$$\left| \sum_{n=1}^{m-1} \lambda(n)\lambda(m-n) \right| < m-1 \quad (m \geq 11). \quad (1)$$

In particular, some pair summing to m has equal Liouville values: otherwise every term would be -1 , making the absolute value exactly $m-1$. The descent below turns this unspecified equal-sign pair at m into a pair of negative signs at $4m$.

A descent

Lemma. Let $m > 1$ be odd and not divisible by three. If no pair of positive integers summing to $4m$ has both Liouville values -1 , then

$$\lambda(n) = -\lambda(m-n) \quad (1 \leq n < m).$$

Proof. Multiplication by four preserves Liouville values, so there is no pair of negative signs summing to m . Multiplication by two reverses them, so there is no pair of positive signs summing to $2m$.

It remains to exclude a pair $a + b = m$ with $\lambda(a) = \lambda(b) = 1$. Suppose such a pair exists. Since m is odd, we may order each such pair with $a < b$. Choose one minimizing $b - a$.

First, neither summand is divisible by three. If $b = 3c$, then $\lambda(c) = -1$. The absence of a pair of negative signs at m forces $\lambda(m-c) = 1$, and hence

$$\lambda(3m-b) = \lambda(3(m-c)) = -1.$$

The complement of $3m-b$ at $4m$ is $m+b$, so $\lambda(m+b) = 1$. But $a + (m+b) = 2m$ is now a forbidden pair of positive signs. Interchanging a and b also excludes divisibility of a by three.

Since 3 does not divide $m = a + b$, the nonzero residues of a and b modulo three must be equal. Therefore

$$a' = \frac{m+a}{3}, \quad b' = \frac{m+b}{3}$$

are positive integers with $a' + b' = m$. The pairs $b + (m + a) = 2m$ and $a + (m + b) = 2m$ force $\lambda(m + a) = \lambda(m + b) = -1$. Multiplication by three reverses Liouville values, so $\lambda(a') = \lambda(b') = 1$. Yet

$$0 < b' - a' = \frac{b-a}{3} < b - a,$$

contradicting minimality. Thus every pair at m has opposite signs, as claimed. $\square$

Completing the proof

Write $N = 4m$. Suppose first that $m \geq 11$ and neither two nor three divides m . If the required pair at $4m$ were absent, the lemma would force every summand in (1) to be -1 , contradicting Mangerel's bound.

Next observe that

$$8 = 3 + 5 = 4 + 4, \quad 12 = 5 + 7 = 6 + 6.$$

For each total, the first pair has two negative signs and the second has two positive signs. Multiplying all summands by any positive integer d multiplies both signs by $\lambda(d)$, so these two possibilities are either preserved or interchanged. Hence every multiple of eight or twelve has a pair of negative signs. This covers $2 \mid m$ and $3 \mid m$.

The only positive integers $m < 11$ divisible by neither two nor three are 1, 5, 7. They are covered by

$$4 = 2 + 2, \quad 20 = 3 + 17, \quad 28 = 5 + 23,$$

whose summands are all prime. This proves the theorem. $\square$

Formal verification. The accompanying Lean 4 development checks the elementary argument and the deduction from Mangerel's bound, using Mathlib's Liouville function. Mangerel's published theorem is an explicit input; its proof is not part of the formalization. The mathematical theorem above is unconditional.

References

1. M. Shusterman, *Goldbach's conjecture for the Liouville function*, MathOverflow question 307479, 3 August 2018. <https://mathoverflow.net/q/307479>
2. A. P. Mangerel, *On a Goldbach-Type Problem for the Liouville Function*, International Mathematics Research Notices **2024**, no. 16, 11865-11877. <https://doi.org/10.1093/imrn/rnae149>
3. A. P. Mangerel, *On Shusterman's Goldbach-type problem for sign patterns of the Liouville function*, arXiv:2412.17199, 2024. <https://arxiv.org/abs/2412.17199>